BIO203 FINALTERM MEGA FILE

ALL DATA FROM PAST PAPERS (2017-2022)

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SHORT QUESTIONS

Qno1: Difference between Blotting Technique on basis of detection.

SOUTHERN BLOTTING:

The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

NORTHERN BLOTTING:

Northern blot is a method used for detection of a specific RNA sequence in RNA samples.

WESTERN BLOTTING:

A technique used to detect the presence of a specific protein in a complex protein mixture.

Qno2: Write down the limitations of Western blotting?

- A minute imbalance at any level blotting of the procedure can skew the results of the entire process.
- Incorrect labeling of the protein can happen due to the reaction of secondary antibody.
- erroneous in bands or no bands due to insufficient transfer.
- Well trained technicians are required for this technique.
- Primary antibody availability is crucial.
- It is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein are possible.

Qno3: Nomenclature of enzyme.

A suitable system was proposed by Smith and Nathans (1973).

- The species name of the organisms is identified by the first letter of genus and first two letters of specific epithet like E.coli = Ecoli.
- Strain identification is written as EcoK.
- In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of H. influenzae. Like HindI, HindII, HindIII etc.
- All restriction enzymes have general name endonuclease R and modification-methylase M. Followed by the system name, for example, in case of H. influenzae
- R. HindIII or M. HindIII.

Qno4. Write properties of probes?

PROBES

- Labeled material to detect a target.
- For DNA: 20-30 nucleotides
- complementary to a region in the gene or DNA.

Qno5: Write applications of PCR in clinical microbiology.

PCR-Medicine

- PCR has major impact on medicine especially in the field of clinical microbiology or diagnosis.
- Molecular tools have also allowed performing prenatal genetic diagnosis.

Qno6: Explain R-M system.

Types of restriction and modification (R-M) system

At least 4 R-M system are known.

- Type-I
- Type-II
- Type-III
- Type-IIs

Type I

Type I were the First to be characterized from E. coli K12.The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit. Type I are of little value for gene manipulation.

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Type II

Most of the useful R-M system is Type II. Type II R-M system recognize a defined sequence and cut within it.

Type III

Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value.

Type IIs

Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness.

Qno7: Define SNP.

A SNP is defined as a single base change in a DNA sequence that occurs in a significant proportion (more than 1 percent) of a large population.

Qno8: Write down method of DNA Finger printing.

Types of DNA fingerprinting methods are

• Electrophoresis.

• Polymerase chain reaction (PCR).

• Restriction fragment length polymorphism (RFLP).

• Random Amplified Polymorphic DNA (RAPD).

• Amplified fragment length polymorphism (AFLP).

Qno9: Explain Western Blotting.

WESTERN BLOTTING

A technique used to detect the presence of a specific protein in a complex protein mixture.

Procedure

- To determine the molecular weight of a protein.
- To measure relative amounts (quantitation) of the protein present in complex mixtures of proteins that are not radiolabeled.
- Western blots have become one of the most common analytical tools for the detection of viral proteins.
- Characterization of monoclonal and polyclonal antibody preparations and in determining the specificity of the immune response to viral antigens.

Qno10: When does RFLP occur?

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An RFLP occurs when the length of a detected fragment varies between individuals. Each fragment length is considered an allele, and can be used in genetic analysis.

Qno11: Write down benefits is secondary antibodies in western blotting.

A species-specific, labeled secondary antibody directed against the constant region of the primary antibody is then used. The most common antibody label used in Western blots is HRP.

Qno12: Write down the Methods of joining DNA fragment.

Methods of joining DNA fragments:

Mainly three methods are used for joining DNA in vitro

- Joining covalently annealed cohesive ends by DNA ligase
- Joining blunt-ended fragments by DNA ligase from phage T4infected E.coli.
- The third utilizes the enzyme terminal deoxynucleotidyl transferase to synthesize homopolymeric 3'single stranded tails at the ends of fragments.

Qno13: DNA ligase Activity.

DNA ligase:

E.coli and phage T4 encode enzyme, DNA ligase, which seals single stranded nicks between adjacent **n**ucleotides in a duplex DNA chain. DNA ligase to create covalent recombinant DNA ligase can repair the nicks produced after the association of cohesive ends of DNA strands.

Qno14: When does RFLP occur?

• An RFLP occurs when the length of a detected fragment varies between individuals. Each fragment length is considered an allele, and can be used in genetic analysis.

Qno15: Principle of PCR in clinical Microbiology.

CLINICAL MICROBIOLY

BACTERIA

- Identification of Bacteria
- Slow growing Bacteria
- Bacterial antibiotic resistance genes.
  VIRUS:
- almost all Viruses identification and viral load.

FUNGUS:

- Aspergillus spp.
- Candida spp
- Plasmodium spp
- Trypanosoma spp

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- Leishmania spp
- Babesia

Qno16: Benefits is secondary antibodies in Western blot

Protein of interest is detected and localized using a specific antibody. Western blotting Protocols utilize a non-labeled primary antibody directed against the target protein. A species-specific, labeled secondary antibody directed against the constant region of the primary antibody is then used. The most common antibody label used in Western blots is HRP. The signal is detected when HRP is exposed to A substrate solution in the final step of the Immunodetection procedure.

Qno17: Characteristics of SNPs

These variations can be:

- Harmless (change in phenotype)

Harmful (diabetes, cancer, heart disease, Huntington, and hemophilia)

Latent variations found in coding and regulatory regions, are not harmful on their own, and the change in each gene only becomes apparent under certain conditions e.g. Susceptibility to lung cancer)

- SNPs are found in coding and (mostly) noncoding regions.
- Occur with a very high frequency about 1 in 1000 bases to 1 in 100 to 300 bases.
- The abundance of SNPs and the ease with which they can be measured make these genetic variations significant.
- SNPs close to particular gene acts as a marker for that genes.
- SNPs in coding regions may alter the protein structure made by that coding region.

Qno18: DNA Lligase activity

- DNA ligase to create covalent recombinant DNA.
- DNA ligase can repair the nicks produced after the association of cohesive ends of DNA strands
- Blunt end ligation via linker molecules
- Joining DNA fragments with cohesive ends by DNA ligase is a relatively efficient process

T4DNA ligase has been used to joint blunt-ended DNA molecule.

Qno19: Names of blotting techniques

Southern, northern and western blotting techniques

Qno20: Southern Blotting is used for

Southern blotting technique is used for the detection of DNA.

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Qno21: VNTR

VNTR loci are very similar between closely related humans.

• The analysis of variable number of tandem repeats (VNTRs), to detect the degree of relatedness to another sequence of oligonucleotides, making them ideal for DNA fingerprinting.

- A Variable Number Tandem Repeat (or VNTR) is a location in a genome where a short nucleotide sequence is organized as a tandem repeat.
- These can be found on many chromosomes, and often show variations in length between individual
- Each variant acts as an inherited allele, allowing t hem to be used for personal or parental identification

Qno22: Screening techniques of SNPs

Two different screening strategies

- Many SNPs in a few individuals
- A few SNPs in many individuals

Qno23: SNPs

A SNP is defined as a single base change in a DNA sequence that occurs in a significant proportion (more than 1 percent) of a large population.

Qno24: Method of dna fingerprinting

1) Isolation of DNA :

2) Cutting, sizing, and sorting

3) Transfer of DNA to nylon

4) Probing

5) DNA fingerprint

Qno25: Western Blotting

WESTERN BLOTTING

A technique used to detect the presence of a specific protein in a complex protein mixture.

- To determine the molecular weight of a Protein.
- To measure relative amounts (quantitation) of the protein present in complex mixtures of proteins that are not radiolabeled.

Qno27: Procedure of southern blotting and How it cleavage

PROCEDURE

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- DNA is extracted from cells, leukocytes.
- DNA is cleaved into many fragments by restriction enzyme (e.g, BamH1, EcoR1 etc).
- The resulting fragments are separated on the basis of size by electrophoresis.
- The DNA fragments are denatured and transferred to nitrocellulose membrane for analysis.
- The labeled probe is added to the blocked membrane in buffer and incubated for several hours to allow the probe molecules to find their targets.
- Blot is incubated with wash buffers containing NaCl and detergent to wash away excess probe.
- Radioactive probes enable autoradiography detection.

CONCLUSION

- Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

Qno28: mechanism DNA ligase

DNA ligase

- An enzyme that creates a phosphodiester bond between the 3' end of one DNA segment and the 5' of another.

Qno29: Adaptor and linker difference

The main difference between linker and adaptor is that former having blunt ends while later have one cohesive end

Qno30: Probes

A probe is a single-stranded sequence of DNA or RNA used to search for its complementary sequence in a sample genome. The probe is placed into contact with the sample under conditions that allow the probe sequence to hybridize with its complementary sequence

Qno31: Radioactive and Non-radioactive probe

RADIOACTIVE PROBE – P32

- Sensitive
- Relatively cheap
- Hazardous
- Radioactive waste disposal regulations should be followed

NON-RADIOACTIVE PROBE – BIOTIN

- Sensitive
- Relatively expensive.

Qno32: Mechanism of WGA

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Primers can be designed to be 'degenerate' – Able to initiate replication from a large number of **t**arget locations. Whole genome amplification (WGA) is a group of procedures that allow amplification to occur at **m**any locations in a genome.

Qno33: Limitations of PCR

LIMITATION

- Contamination risk
- Primer complexities
- Primer-binding site complexities
- Amplifies rare species
- Detection methods properties of DNA probe

Qno34: Explain mini primer PCR

MINI PRIMER PCR

Mini Primer PCR uses a thermostable Polymerase (S-Tbr) that can extend from short **p**rimers as short as 9 or 10 nucleotides. This method permits PCR targeting to smaller primer binding regions, and is used to amplify **c**onserved DNA sequences, such as the 16S (or Eukaryotic 18S) rRNA gene. PCR that can extend from short primers.

Qno35: What is the purpose of blank control and negative control in PCR reaction

BLANK REACTION

Controls for contamination contains all reagents except DNA template.

NEGATIVE CONTROL

Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.

Qno36: What is alu-PCR and write it purpose

The use of primers from a commonly repeated segment is called Alu-PCR, and can help to **a**mplify sequences adjacent (or between) these repeats.

Qno37: Mutation identification

MUTATION IDENTIFICATION

- The presence of a mutation affecting a restriction site causes the pattern of bands to differ from those seen in a normal gene.
- A change in one nucleotide may alter the nucleotide sequence so that the restriction endonuclease fails to recognize and cleave at that site.
- Digestion of genomic DNA to DNA fragments.
- Size-separation of the fragments.
- In situ denaturation of the DNA fragments.

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- Transfer of denatured DNA fragments into a solid support (nylon or nitrocellulose).
- Hybridization of the immobilized DNA to a labeled probe (DNA, RNA).
- Detection of the bands complementary to the probe (e.g. by autoradiography).
- Estimation of the size & number of the bands generated after digestion of the genomic DNA will be different.

Qno38: Use of asymmetric PCR

ASYMMETRIC PCR

Asymmetric PCR is used to amplify one strand of the original DNA more than the other. It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal. PCR is carried out as usual, but with a great excess of one primers for the chosen strand. Due to the slow amplification later in the reaction after the limiting primer has been used up, extra cycles of PCR are required. It finds use in some types of sequencing and hybridization **p**robing where having only one of the two complementary strands is required.

Qno39: What is the first step of RT.PCR

REVERSE TRANSCRIPTASE PCR

First step of RT-PCR – first strand reaction Synthesis of cDNA using oligo dT primers (37°C) one hour.

Qno40: Methylation PCR

METHYLATION SPECIFIC PCR

Methylation-specific PCR is used to identify patterns of DNA Methylation at CpG Islands in genomic DNA. Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to Uracil, which is complementary to adenosine in PCR primers. Two amplifications are then carried out on the Bisulfite-treated DNA: One primer set anneals to DNA with cytosine (corresponding to Methylated cytosine), and the other set anneals to DNA with uracil (corresponding to Unmethylated cytosine). MSP used in quantitative PCR provides information about methylation state of a given CpG island.

Qno41: Inverse PCR and conventional PCR difference

Inverse PCR uses Standard PCR primers oriented in the reverse direction **o**f the usual O. The template for the reverse primers is a restriction fragment that has been self-ligated. Inverse PCR functions to clone sequences Flanking a known sequence. Flanking DNA sequences are digested and then ligated to generate circular DNA.

Qno42: What solid support is used in southern blotting and hybridization components

Nylon or nitrocellulose are used as solid supports in southern blotting technique and hybridization components.

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Qno43: Pfu, taq and vent source and extension rate.

Sources

Pfu: Pyrococcus furiosus (fidelity)

Taq: Thermus aquaticus (most commonly used)

Vent: Thermococcus litoralis

Extension rates.

Pfu = 60nt/sec

Taq = 75 nt/sec

Vent = > 80 not/sec

Qno44: Primer designing for whole genome amplification PCR.

Primers can be designed to be 'degenerate' – able to initiate replication from a large number of target locations.

Qno45: What is the use of methylation PCR Write a note on mini PCR Why we use Southern blotting Technique

Methylation-specific PCR is used to identify patterns of DNA Methylation at CpG Islands in genomic DNA.

Mini Primer PCR uses a thermostable Polymerase (S-Tbr) that can extend from short primers as short as 9 or 10 nucleotides. This method permits PCR targeting to smaller primer binding regions, and is used to amplify conserved DNA sequences, such as the 16S (or Eukaryotic 18S) rRNA gene. PCR that can extend from short primers.

Southern blotting technique is used for the detection of DNA.

Qno46: Nested PCR primers.

NESTED PCR

Two pairs (instead of One pair) of PCR Primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR.

NESTED PCR – SECOND PAIR OF PRIMERS

Second pair of primers – nested primers (as they lie / are nested within the

first fragment).

SECOND PCR PRODUCT IS SHORTER THAN FIRST ONE

Second pair of primers bind inside the first PCR product fragment to allow amplification of a second PCR product.

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SECOND PCR PRODUCT IS SPECIFIC

It increases the specificity of DNA amplification, by reducing background due to non-specific amplification of DNA.

ADVANTAGE OF NESTED PCR

Very low probability of nonspecific amplification.

Qno47: First step of PCR

DENATURATION

First regular cycling event and consists of heating the reaction to 93°C – 98°C for 20-45 seconds. It causes melting of DNA template yielding single strands of DNA.

Qno48: Colony PCR

COLONY PCR

The screening of bacterial or yeast clones for correct Ligation or plasmid Products.

COLONY PCR METHODOLOGY

The screening of bacterial or yeast clones for correct ligation or plasmid products. Pick a bacterial colony with an autoclaved toothpick, swirl it into 25 µl of TE autoclaved dH2O in an microfuge tube. Heat the mix in a boiling water bath (90-1000C) for 2 minutes. Spin sample for 2 minutes high speed in centrifuge . Transfer 20 µl of the supernatant into a new microfuge tube. Take 1-2 µl of the supernatant as template in a 25 µl PCR standard PCR reaction. The screening of Bacterial or yeast clones for correct ligation or plasmid products.

Qno49: conventional PCR reagents and their use

MAJOR INGREDIENTS

- Microfuge tube__These are small cylindrical plastic tubes with conical bottom.
- Thermal cycler___ The thermocycler works on the principle of Pettier effect, which raises and lowers the temperature of the block in a pre-programmed manner.
- DNA template___ For denaturation and conversation into 2 single strands
- Primers___ Primers are single stranded 18–30 bp long DNA fragments Complementary to sequences flanking the region to be amplified.
- Buffer___ Provides suitable environment for the activity of DNA polymerase.
- MgCl2___ Acts as cofactor and enhance the activity of polymerases.
- Distilled water___ Free of nucleotides so that the DNA products don't get ddegrade
- DNA polymerase___

DNA polymerase can add nucleotides by using primer.

- Triphosphates

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dNTPs are the essential building blocks of nucleic acid molecules, and as such are necessary components of PCR mixes as no new amplified DNA could be generated without these.

- DNA polymerase___ Play key role in synthesizing new DNA strand.

Qno50: Application of suicide PCR

Suicide PCR is typically used in paleogenetics or other studies where avoiding false positives and ensuring the specificity of the amplified fragment is the highest priority. The method prescribes the use of any primer combination only once in a PCR, which should never have been used in any positive control PCR reaction.

Qno51: Limitations of western blotting

Limitations in western blotting very delicate and time consuming process. A minute Imbalance at any level of the procedure can skew the results of the entire process. Incorrect labeling of the protein can happen due to the reaction of secondary antibody. Cause erroneous in bands or no bands due to insufficient transfer. Well trained technicians are required for this technique. Primary antibody availability is crucial. It is just a semi-quantitative at best. Only an approx. estimation & not a precise measurement of molecular weight of the protein is possible

Qno52: Principal families of VNTRs

There are two principal families of VNTRs:

- **Microsatellites**.
- **Minisatellites**.

Microsatellites, also known as Simple Sequence repeats (SSRs) or short tandem repeats (STRs), are repeating sequences of 2-6 base pairs of DNA.

A minisatellites (also referred as VNTR) is a section of DNA that consists of a short series of bases 10– 60 base pairs. Their analysis is useful in genetics and biology research, forensics, and DNA fingerprinting.

Qno53: Difference between denaturation and extension

DENATURATION

First regular cycling event and consists of heating the reaction to 93°C – 98°C for 20- 45 seconds. It causes melting of DNA template yielding single strands of DNA.

EXTENSION

The temperature at this step depends on the type of DNA polymerase used. Taq polymerase has its optimum activity temperature at 70- 80°C. Commonly 72°C is used.

Qno54: Principle of Electrophoresis

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Electrophoresis is a separations technique that is based on the mobility of ions in an electric field. Positively charged ions migrate towards a negative electrode and negatively- charged ions migrate toward a positive electrode. Ions have different migration rates depending on their total charge, size, and shape, and can therefore be separated.

Qno55: Describe requirements for primer designing

PRIMERS FOR PCR

- Avoid inter-strand homologies
- Avoid intra-strand homologies
- Tm of forward primer = Tm of reverse primer
- G/C content of 20–80%; avoid longer than GGGG
- Product size (100–700 bp)
- Target specificity

Qno56: Define microscope magnification

Magnification is the ability of a microscope to produce an image of an object at a scale larger (or even smaller) than its actual size. Magnification serves a useful purpose only when it is possible to see more details of an object in the image than when observing the object with the unaided eye.

Qno57: Mispriming

Misprimes may occur due to non-specific hybridization of primers.

Qno58: Primer dimer

- Primers dimers may reduce amplification
- Primer dimers may occur due to hybridization of primers to each other.

Qno59: Long PCR

LONG PCR

Extended or longer than standard PCR, meaning over 5 Kilobases (frequently over 10 kb). Long PCR is useful only if it is accurate.

MIXTURES OF POLYMERASES

Special mixtures of proficient polymerases along with accurate polymerases such as Pfu are often mixed together.

Application – to clone **l**arge genes not possible with conventional PCR.

Qno60: Multiplex PCR

MULTIPLEX PCR

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Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers.

AMPLICONS OF DIFFERENT SIZES ARE PRODUCED

Multiplex-PCR consists of multiple primer sets within a single PCR mixture to produce amplicons of varying sizes that are specific to different DNA sequences.

SINGLE REACTION, MANY AMPLICONS

By targeting multiple sequences at once, additional information may be gained from a single test run that otherwise would require several times the reagents and more time to perform.

ANNEALING TEMPERATURES BE OPTIMIZED

Annealing temperatures for each of the primer sets must be optimized to work correctly within a single reaction.

AMPLICON SIZE SHOULD BE DIFFERENT OR BE LABELED

Amplicon sizes should be different enough to form distinct bands when visualized by gel electrophoresis.

LABELED PRIMERS

If amplicon sizes overlap, the different amplicons may be differentiated and visualized using primers that have been dyed with different color fluorescent dyes.

PRIMERS PAIRS BE OPTIMIZED

The primer design for all primers pairs has to be optimized so that all primer pairs can work at the same annealing temperature during PCR.

MULTIPLEX PCR

The use of multiple, Unique primer sets **w**ithin a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences.

Qno61: Stages of PCR

1- Denaturation
2- Annealing
3- Extension

Qno62: Write chemical agents of Radioactive Probe

Radioactive Probe – P32

Qno63: Direct sequencing

Sanger dideoxy sequencing can detect any type of unknown polymorphism and its position, when the majority of DNA contains that polymorphism. Misses polymorphisms and mutations

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when the DNA is heterozygous**.** The limited utility for analysis of solid tumors or pooled **s**amples of DNA due to low sensitivity. Once a sample is known to contain a polymorphism in a specific region, direct sequencing is particularly useful for identifying a polymorphism and its specific position. Even if the identity of the polymorphism cannot be discerned in the first pass, multiple sequencing attempts have proven quite successful in elucidating sequence and position information.

Qno64: Difference b/w Mini satellite and micro satellite.

Microsatellites, also known as Simple Sequence Repeats (SSRs) or short tandem repeats (STRs), are repeating sequences of 2-6 base pairs of DNA. A minisatellites (also referred as VNTR) is a section of DNA that consists of a short series of bases 10– 60 base pairs.

Qno64: Write the principle of thermocycler

THERMOCYCLER

The thermocycler works on the principle of Peltier effect, which raises and lowers the temperature of the block in a pre-programmed manner.

Qno65: Which enzyme is use in mini PCR

Mini Primer PCR uses a thermostable Polymerase (S-Tbr) that can extend from short Primers as short as 9 or 10 nucleotides.

Qno66: Advantage of PCR

ADVANTAGES

- Specific
- Simple
- Rapid
- Relatively inexpensive
- Amplifies from low quantities
- Works on damaged DNA
- Sensitive
- Flexible

Qno67: what is in situ? Write principal and uses

IN SITU PCR

In Situ PCR (ISH) is a polymerase chain reaction that actually takes place inside the cell on a slide.

In situ PCR **a**mplification can be performed on fixed tissue or cells .Applies the methodology of hybridization of the nucleic acids. Allows identification of cellular markers. Limited to detection of non-genomic material such as RNA.

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Qno68: Name the method we use transfer of DNA to membrane

In southern blotting technique The DNA fragments are denatured and transferred to nitrocellulose membrane for analysis.

Qno69: write the application of western blotting

Applications of western blotting

- Analysis of IgG Fractions purified from human plasma.
- Diagnosis of HIV by ELISA, involves the western blotting technique.
- Western blotting technique is also used to Detect Some Forms of Lyme Disease.
- Western blotting technique is used in Definitive Test for BSE, which is commonly known as Mad cow disease.
- Confirmatory Test for Hepatitis-B involves western blotting technique.

Qno70: Difference b/w known polymorphism and unknown polymorphism techniques

Known polymorphism

Hybridization Techniques

- Micro arrays
- Real time PCR
- Enzyme based Techniques
- Nucleotide extension
- Cleavage
- Ligation
- Reaction product detection and display

Unknown polymorphism

- Direct Sequencing
- Microarray
- Cleavage / Ligation
- Electrophoretic mobility assays

Qno71: Temperature effect on primers

ANNEALING TEMPERATURES BE OPTIMIZED

Annealing temperatures for each of the primer sets must be optimized to work correctly within a single reaction.

Qno72: Palindrome

Type II endonucleases recognize and cleave DNA within particular sequences of four to eight nucleotides which have a twofold axis of rotational symmetry i.e. Referred as palindromes

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5’-GAATTC -3’ 5’-CTTAAG

Qno73: Abbreviations of RFLP, AFLP

RFLP- restriction fragment length polymorphism

AFLP- amplified fragment length polymorphism

Qno74: Types of PCR

TYPES

- Nested PCR
- Multiplex PCR
- Touchdown PCR
- Sequence-specific PCR
- Reverse- Transcriptase PCR
- Long-range PCR
- Whole-genome amplification
- RAPD PCR (AP-PCR)
- Quantitative real-time PCR

LONG QUESTIONS

#1: Transfer methods of DNA to membrane

TRANSFER METHODS OF DNA TO MEMBRANE

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

#2: Restriction and modification system

Types of restriction and modification (R-M) system

At least 4 R-M system are known.

Type-I

Type-II

Type-III

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Type-IIs

Type I

Type I were the First to be characterized from E. coli K12.The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit. Type I are of little value for gene manipulation**.**

Type II

Most of the useful R-M system is Type II. Type II R-M system recognize a defined sequence and cut within it.

Type III

Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value.

Type IIs

Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness**.**

#3 DNA fingerprinting

The process of DNA fingerprinting was developed by Professor Alec Jeffrey at Leicester University in 1984 as a form of genetic analysis. • It was first used in the law courts of England in 1987 to convict a man in a rape case.

• It has now been used successfully in many crime and paternity cases in worldwide. Although 99.9% of human DNA sequences are the same in every person, enough of the DNA is different to distinguish one individual from another, unless they are monozygotic twins.

• DNA profiling uses repetitive sequences that are highly variable, called variable number tandem repeats (VNTRs), particularly short tandem repeats. (STRs). VNTR loci are very similar between closely related humans.

• The analysis of variable number of tandem Repeats (VNTRs), to detect the degree of relatedness to another sequence of oligonucleotides , making them ideal for DNA fingerprinting. A Variable Number Tandem Repeat (or VNTR) is a location in a genome where a short nucleotide sequence is organized as a tandem Repeat. These can be found on many chromosomes, and often show variations in length between individuals.Each variant acts as an inherited allele, allowing t hem to be used for personal or parental identification.

- There are two principal families of VNTRs:
- Microsatellites.
- Minisatellites

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Microsatellites, also known as Simple Sequence repeats (SSRs) or short tandem repeats (STRs), are repeating sequences of 2-6 base pairs of DNA.

A minisatellites (also referred as VNTR) is a section of DNA that consists of a short series of bases 10– 60 base pair. Their analysis is useful in genetics and biology research, forensics, and DNA fingerprinting.

#4: Calculate the four, six, eight restriction fragments system

Number and size of Restriction fragments

The number and size of **f**ragments generated by restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut.

- Four base recognition site occurs every 4 power 4 (256) bp.
- Six base recognition site occurs every 4 power of 6 (4096) bp.
- Eight base recognition site occurs 4 power of 8 (65,536)bp.

#5: haplotype

The "Haplotype centric" approach combines the information of adjacent SNPs into composite multilocus haplotypes. Haplotypes are not only more informative but also capture the regional LD information, which is assumed to be robust and powerful association of haplotype frequencies with the presence of desired phenotypic frequencies in the population will help in utilizing the maximum potential of SNP as a marker.

#6: western blotting procedure

Procedure

1- Sample preparation
2- Gel Electrophoresis
3- Blotting (or transfer)
4- Blocking
5- Antibody probing
6- Detection

1- The choice of extraction method depends primarily on the sample and whether the analysis is targeting all the proteins in a cell or only a component from a particular subcellular fraction.
2- Samples are loaded into separate wells. A protein marker is also loaded. The separated protein mixtures are transferred to a solid support for further analysis.
3- Transfer can be done in wet or semi-dry conditions. Semi-dry transfer is generally faster. Wet transfer is recommended for large proteins, >100 kD.
4- Blocking is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blotting membrane.

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5- Protein of interest is detected and localized using a specific antibody. Western blotting protocols utilize a non-labeled primary antibody directed against the target protein. A species-specific, labeled secondary antibody directed against the constant region of the primary antibody is then used. The most common antibody label used in Western blots is HRP.
6- The signal is detected when HRP is exposed to a substrate solution in the final step of the Immunodetection procedure.

#7: steps of fingerprinting

1)Isolation of DNA:

DNA must be recovered from the cells or tissues of the body. Only a small amount of tissue, like blood, hair, or skin, is needed. For example, the amount of DNA found at the root of one hair is usually sufficient.

2) Cutting, sizing, and storage:

Special enzymes called restriction enzymes are used to cut the DNA at specific sites. For example, an enzyme called EcoR1, found in bacteria, will cut DNA only when the sequence 5'..GAATTC..3' occurs. The DNA pieces are sorted according to size by a sieving technique called electrophoresis. The DNA pieces are passed through a gel agarose. This technique is the DNA equivalent of screening sand through progressively finer mesh screens to determine particle sizes.

3) Transfer of DNA to nylon:

The distribution of DNA pieces is transferred to a nylon sheet by placing the sheet on the gel and soaking them overnight.

4) Probing :

Adding radioactive or colored probes to the nylon sheet produces a pattern called the DNA fingerprint. Each probe typically sticks in only one or two specific places on the nylon sheet.

5) DNA fingerprint :

The final DNA fingerprint is built by using several probes (5-10 or more) simultaneously. It resembles the bar codes used by grocery store scanners**.**

#8: Methods to transfer DNA on membrane and procedure for upward transfer of DNA

TRANSFER METHODS OF DNA TO MEMBRANE

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

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#9: DNA fingerprinting in diagnosing and developing cure inherit disorders

Diagnosis and developing cures for Inherited disorders:

DNA fingerprinting is used to diagnose inherited disorders in both prenatal and newborn babies in hospitals around the world. These disorders may include cystic fibrosis, Hemophilia, Huntington's disease, familial Alzheimer's, sickle cell anemia, thalassemia, and many others. Early detection of such disorders enables the medical staff to prepare themselves and the parents for proper treatment of the child. In some programs, genetic counselors use DNA fingerprint information to help prospective parents. In other programs, prospective parents use DNA fingerprint information in their decisions concerning affected pregnancies.

#10: PCR and it's types

PCR

A technique widely used in Molecular biology and biotechnology. Its name is from one of its key component – DNA polymerase. As PCR progresses, DNA generated is itself used as template for replication. This sets in motion of a chain reaction in which the DNA template is exponentially amplified. During PCR, a heat stable DNA polymerase, such as Taq polymerase – an enzyme derived from the bacterium Thermus Aquatics.

THREE STEPS

- Denaturation
- Annealing
- Extension

INITIALIZATION

This step consists of heating the reaction to a temperature of 93°C – 96°C, which is held for 1-10 minutes.

DENATURATION

First regular cycling event and consists of heating the reaction to 93°C – 98°C for 20- 45 seconds. It causes melting of DNA template yielding single strands of DNA.

ANNEALING

The reaction temperature is lowered to 40-65°C for 20-40 seconds. Annealing of the primers to the single stranded DNA template.

EXTENSION

The temperature at this step depends on the type of DNA Polymerase used. Taq polymerase has its optimum activity temperature at 70- 80°C. Commonly 72°C is used. At this step the DNA Polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding dNTP's.

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FINAL ELONGATION

Single step is performed at a temperature of 70- 74°C for 5-15 minutes after the last PCR cycle to ensure that any remaining single-stranded DNA is fully extended.

FINAL HOLD TEMPERATURE

This step at 4-15°C for an indefinite time may be employed for short-term storage of the reaction.

#11: blotting techniques

SOUTHERN BLOTTING:

The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

NORTHERN BLOTTING:

Northern blot is a method used for detection of a specific RNA sequence in RNA samples.

WESTERN BLOTTING:

A technique used to detect the presence of a specific protein in a complex protein mixture.

#12: southern blotting in detail

SOUTHERN BLOTTING

The Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe. Southern Blotting is named after its inventor, the British Biologist Edwin Southern (1975). Identify DNA sequence (gene) of interest. Identified DNA may be a small piece of DNA or a mutation.

PROCEDURE

- DNA is extracted from cells, leukocytes.
- DNA is cleaved into many fragments by restriction enzyme (e.g, BamH1, EcoR1 etc)
- The resulting fragments are separated on the basis of size by electrophoresis.
- The DNA fragments are denatured and transferred to nitrocellulose membrane for analysis.
- The labeled probe is added to the blocked
- Membrane in buffer and incubated for several
- Hours to allow the probe molecules to find their targets.
- Blot is incubated with wash buffers containing
- NaCl and detergent to wash away excess probe.
- Radioactive probes enable autoradiographic detection.

CONCLUSION

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Southern blot is used to detect the presence of a particular piece of DNA in a sample by a molecular probe.

PROBES

Labeled material to detect a target. For DNA: 20-30 nucleotides, complementary to a region in the gene or DNA.

RADIOACTIVE PROBE – P32

Sensitive, Relatively cheap, Hazardous, Radioactive waste disposal regulations should be followed

NON-RADIOACTIVE PROBE – BIOTIN

Sensitive, Relatively expensive

HYBRIDIZATION OF PROBES

The binding between single stranded labeled probe to a complementary nucleotide sequence on the target DNA.

PROBES

Labeled material to detect complementary region in the gene or DNA.

TRANSFER METHODS OF DNA TO MEMBRANE

- Upward capillary transfer
- Downward capillary transfer
- Simultaneous transfer to two membranes
- Electrophoretic transfer
- Vacuum transfer

#13: northern blotting in detail

NORTHERN BLOTTING

Northern blot is a **m**ethod used for detection of a specific RNA sequence in RNA samples**.**

PROCEDURE

- Isolation of intact mRNA.
- Separation of RNA according to size (through **a** denaturing agarose gel.
- Transfer of the RNA to a solid support.
- Fixation of the RNA.
- Hybridization of the immobilized RNA to probes complementary to the sequences of interest.
- Removal of probe molecules that are nonspecifically bound to the solid matrix.
- Detection, capture and analysis of an image of the specifically bound probe molecules.

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- Northern blot is a method used for detection of a specific RNA sequence in RNA samples.

APPLICATIONS

Study of gene expression in eukaryotic cells.

- To measure the amount & size of RNAs transcribed from eukaryotic genes.
- To estimate the abundance of RNAs.
- To equalize the amounts of RNA loaded into Lanes of gels.
- Use of housekeeping gene (endogenous Constitutively-expressed gene).
- Normalizing samples according to their content of mRNAs of the housekeeping gene.

#14: western blotting technique

WESTERN BLOTTING

A technique used to **d**etect the presence of a specific protein in a complex protein mixture. To determine the molecular weight of a **p**rotein. To measure relative amounts (quantitation) of the protein present in complex mixtures of proteins that are not radiolabeled**.** Western blots have become one of the most common analytical tools for the detection of Viral proteins. Characterization of monoclonal and polyclonal antibody preparations and in determining the specificity of the immune response to viral antigens. A technique used to detect the presence of a specific protein in a complex protein mixture.

PROCEDURE

- Sample preparation
- Gel Electrophoresis
- Blotting (or transfer)
- Blocking
- Antibody probing
- Detection

Procedure

- The choice of extraction method depends primarily on the sample and whether the analysis is targeting all the proteins in a cell or only a component from a particular subcellular fraction.
- Samples are loaded into separate wells. A protein marker is also loaded. The separated protein mixtures are transferred to a solid support for further analysis.
- Transfer can be done in wet or semi-dry conditions. Semi-dry transfer is generally faster. Wet transfer is recommended for large proteins, >100 kD. Blocking is a very important step in the immunodetection phase of Western blotting because it prevents non-specific binding of antibody to the blotting membrane.

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- Protein of interest is detected and localized using a specific antibody. Western blotting protocols utilize a non-labeled primary antibody directed against the target protein.
- A species-specific, labeled secondary antibody directed against the constant region of the primary antibody is then used. The most common antibody label used in Western blots is HRP.
- The signal is detected when HRP is exposed to a substrate solution in the final step of the Immunodetection procedure.

#15: hot start PCR

HOT START PCR

This is a technique that reduces nonspecific amplification during the initial set up stages of the PCR. The technique may be performed manually by heating the reaction components upto the melting temperature (e.g, 95°C). Before adding the polymerase, specialized enzyme systems have been developed that inhibit the polymerase's activity at ambient temperature. This is done either by the binding of an antibody or by the presence of covalently bound inhibitors that only dissociate after a high-temperature activation step. DNA Polymerase- Eubacterial type I DNA polymerase, Pfu. These thermophilic DNA polymerases show a very small polymerase activity at room temperature.

#16: DNA modifying enzymes

DNA modifying Enzymes

- Nucleases
- DNA Polymerase
- Reverse transcriptase
- DNA ligases

Nucleases

Nucleases or DNases are the enzymes that degrade DNA. Two broad classes of nucleases

i). Exonucleases

ii). Endonucleases

DNA Polymerase

An enzyme that catalyze template-dependent synthesis of DNA.

Reverse transcriptase.

An RNA-directed DNA Polymerase in retroviruses; capable of making DNA complementary to an RNA.

DNA ligase

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An enzyme that creates a Phosphodiester bond between the 3' end of one DNA segment and the 5' of another.

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